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TROUBLESHOOTING FABRIC APPLICATIONS

AIM

To identify, analyze, and resolve common issues encountered in Hyperledger Fabric applications by troubleshooting network connectivity, chaincode deployment, channel configuration, certificate authentication, and Docker container errors.

PROCEDURE

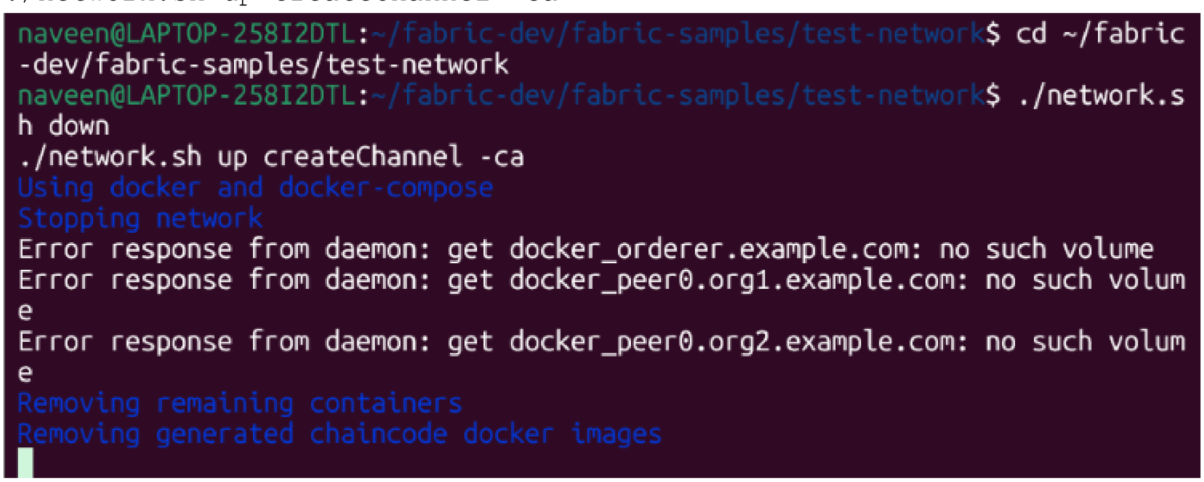
Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh down

./network.sh up createChannel -ca
```



```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ cd ~/fabric
-dev/fabric-samples/test-network
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.s
h down
./network.sh up createChannel -ca
Using docker and docker-compose
Stopping network
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volum
e
Error response from daemon: get docker_peer0.org2.example.com: no such volum
e
Removing remaining containers
Removing generated chaincode docker images
```

Step 3: Verify Docker Containers

Check whether all required Fabric containers are running.

```
docker ps
```

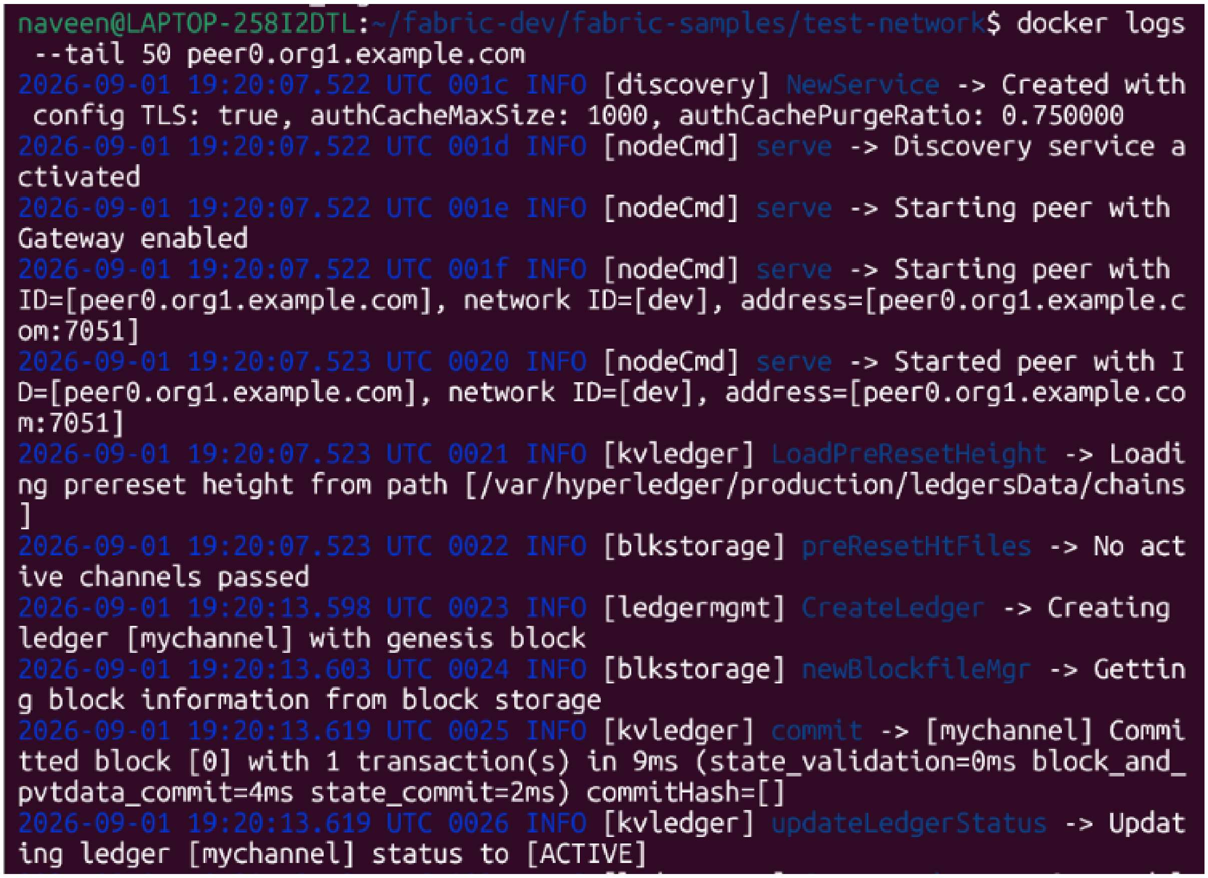
```
Channel 'mychannel' joined
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID   IMAGE                               COMMAND
CREATED            STATUS          PORTS

                   NAMES
39d90c7894e7   hyperledger/fabric-peer:latest      "peer node start"
20 seconds ago     Up 19 seconds   0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp
, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
                   peer0.org2.example.com
2c7ab4c6d7b6   hyperledger/fabric-orderer:latest   "orderer"
20 seconds ago     Up 19 seconds   0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp
, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:
9443->9443/tcp     orderer.example.com
bae49aec171b   hyperledger/fabric-peer:latest      "peer node start"
20 seconds ago     Up 19 seconds   0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp
, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
                   peer0.org1.example.com
b551bd1174c6   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"
26 seconds ago     Up 26 seconds   0.0.0.0:8054->8054/tcp, [::]:8054->8054/tcp
, 0.0.0.0:18054->18054/tcp, [::]:18054->18054/tcp
                   ca_org2
af99b8cf271b   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"
26 seconds ago     Up 26 seconds   0.0.0.0:9054->9054/tcp, [::]:9054->9054/tcp
, 0.0.0.0:19054->19054/tcp, [::]:19054->19054/tcp
                   ca_orderer
c66f0658cae1   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"
26 seconds ago     Up 26 seconds   0.0.0.0:7054->7054/tcp, [::]:7054->7054/tcp
, 0.0.0.0:17054->17054/tcp, [::]:17054->17054/tcp
```

Step 4: Check Peer Logs

Display recent logs from the Org1 peer.

```
docker logs --tail 50 peer0.org1.example.com
```



```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ docker logs
 --tail 50 peer0.org1.example.com
2026-09-01 19:20:07.522 UTC 001c INFO [discovery] NewService -> Created with
 config TLS: true, authCacheMaxSize: 1000, authCachePurgeRatio: 0.750000
2026-09-01 19:20:07.522 UTC 001d INFO [nodeCmd] serve -> Discovery service a
ctivated
2026-09-01 19:20:07.522 UTC 001e INFO [nodeCmd] serve -> Starting peer with
Gateway enabled
2026-09-01 19:20:07.522 UTC 001f INFO [nodeCmd] serve -> Starting peer with
ID=[peer0.org1.example.com], network ID=[dev], address=[peer0.org1.example.c
om:7051]
2026-09-01 19:20:07.523 UTC 0020 INFO [nodeCmd] serve -> Started peer with I
D=[peer0.org1.example.com], network ID=[dev], address=[peer0.org1.example.co
m:7051]
2026-09-01 19:20:07.523 UTC 0021 INFO [kvledger] LoadPreResetHeight -> Loadi
ng prereset height from path [/var/hyperledger/production/ledgersData/chains
]
2026-09-01 19:20:07.523 UTC 0022 INFO [blkstorage] preResetHtFiles -> No act
ive channels passed
2026-09-01 19:20:13.598 UTC 0023 INFO [ledgermgmt] CreateLedger -> Creating
ledger [mychannel] with genesis block
2026-09-01 19:20:13.603 UTC 0024 INFO [blkstorage] newBlockfileMgr -> Gettin
g block information from block storage
2026-09-01 19:20:13.619 UTC 0025 INFO [kvledger] commit -> [mychannel] Commi
tted block [0] with 1 transaction(s) in 9ms (state_validation=0ms block_and_
pvtdata_commit=4ms state_commit=2ms) commitHash=[]
2026-09-01 19:20:13.619 UTC 0026 INFO [kvledger] updateLedgerStatus -> Updat
ing ledger [mychannel] status to [ACTIVE]
```

Step 5: Check Orderer Logs

Display recent logs from the ordering service.

```
docker logs --tail 50 orderer.example.com
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ docker logs
 --tail 50 orderer.example.com
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
2026-09-01 19:20:07.338 UTC 000f INFO [orderer.common.server] Main -> Beginn
ing to serve requests
2026-09-01 19:20:10.470 UTC 0010 INFO [blkstorage] newBlockfileMgr -> Gettin
g block information from block storage
2026-09-01 19:20:10.491 UTC 0011 INFO [orderer.consensus.etcdraft] HandleCha
in -> EvictionSuspicion not set, defaulting to 10m0s
2026-09-01 19:20:10.491 UTC 0012 INFO [orderer.consensus.etcdraft] HandleCha
in -> Without system channel: after eviction Registrar.SwitchToFollower will
 be called
2026-09-01 19:20:10.491 UTC 0013 INFO [orderer.consensus.etcdraft] createOrR
eadWAL -> No WAL data found, creating new WAL at path '/var/hyperledger/prod
uction/orderer/etcdraft/wal/mychannel' channel=mychannel node=1
2026-09-01 19:20:10.499 UTC 0014 INFO [orderer.commmon.multichannel] createA
sMember -> Joining channel: {mychannel  consenter active 1}
2026-09-01 19:20:10.502 UTC 0015 INFO [orderer.consensus.etcdraft] Start ->
Starting Raft node channel=mychannel node=1
2026-09-01 19:20:10.502 UTC 0016 INFO [orderer.common.cluster] Configure ->
Entering, channel: mychannel, nodes: []
2026-09-01 19:20:10.502 UTC 0017 INFO [orderer.common.cluster] Configure ->
Exiting
2026-09-01 19:20:10.502 UTC 0018 INFO [orderer.consensus.etcdraft] start ->
```

Step 6: Verify Channel Membership

Load the Org1 environment if required. Then list the channels joined by the peer.

```
source setOrg1.sh

peer channel list
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ source setO
rg1.sh
peer channel list
-bash: setOrg1.sh: No such file or directory
2026-09-01 19:22:28.189 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorse
r and orderer connections initialized
Channels peers has joined:
mychannel
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

Step 7: Verify Installed Chaincode

Check the chaincode packages installed on the peer.

```
peer lifecycle chaincode queryinstalled
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer lifecy
cle chaincode queryinstalled
Installed chaincodes on peer:
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

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Step 8: Troubleshoot Chaincode Deployment

If the chaincode has not been installed or committed correctly, redeploy it.

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-
basic/chaincode-javascript
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.s
h deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode
-javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-bas
ic/chaincode-javascript --lang node --label basic_1.0
```

Step 9: Verify Chaincode Execution

Query the ledger to verify that the chaincode is functioning correctly.

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chainc
ode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

Step 10: Check Docker Resource Usage

Monitor CPU and memory usage of the Fabric containers.

```
docker stats
```

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docker stats --no-stream

```
1f142960f24d   dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c3
07bc2b5fbe664c87b24c035bee15cbcc7f59b629   0.00%     51.59MiB / 2GiB       2
.52%     369kB / 22.6kB    0B / 4.1kB       24
476d906635bd   dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c3
07bc2b5fbe664c87b24c035bee15cbcc7f59b629   0.00%     50.52MiB / 2GiB       2
.47%     380kB / 23kB      0B / 4.1kB       24
39d90c7894e7   peer0.org2.example.com
                                           3.56%     97.5MiB / 7.479GiB    1
.27%     534kB / 412kB     69.6kB / 537kB   14
2c7ab4c6d7b6   orderer.example.com
                                           0.16%     18.75MiB / 7.479GiB   0
.24%     84.1kB / 257kB    35.7MB / 266kB   13
bae49aec171b   peer0.org1.example.com
                                           3.66%     98.84MiB / 7.479GiB   1
.29%     551kB / 431kB     6.93MB / 537kB   15
b551bd1174c6   ca_org2
                                           0.00%     9.602MiB / 7.479GiB   0
.13%     29.7kB / 44.2kB   3.31MB / 737kB   11
af99b8cf271b   ca_orderer
                                           0.00%     9.078MiB / 7.479GiB   0
.12%     52.9kB / 84.7kB   27MB / 1.05MB    11
c66f0658cae1   ca_org1
                                           0.00%     8.965MiB / 7.479GiB   0
.12%     32.1kB / 49.9kB   20.5kB / 737kB   11
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ docker stat
s --no-streamam
CONTAINER ID   NAME
                                           CPU %     MEM USAGE / LIMIT     M
EM %     NET I/O           BLOCK I/O        PIDS
1f142960f24d   dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c3
07bc2b5fbe664c87b24c035bee15cbcc7f59b629   0.00%     50.75MiB / 2GiB       2
.48%     372kB / 24.7kB    0B / 4.1kB       24
476d906635bd   dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c3
07bc2b5fbe664c87b24c035bee15cbcc7f59b629   0.00%     50.78MiB / 2GiB       2
.48%     382kB / 25.1kB    0B / 4.1kB       24
39d90c7894e7   peer0.org2.example.com
                                           3.27%     97.34MiB / 7.479GiB   1
.27%     559kB / 438kB     69.6kB / 537kB   14
2c7ab4c6d7b6   orderer.example.com
                                           1.54%     19.5MiB / 7.479GiB    0
.25%     84.7kB / 257kB    35.7MB / 266kB   13
bae49aec171b   peer0.org1.example.com
                                           3.18%     99.15MiB / 7.479GiB   1
.29%     577kB / 457kB     6.93MB / 537kB   15
b551bd1174c6   ca_org2
                                           0.00%     9.602MiB / 7.479GiB   0
.13%     29.7kB / 44.2kB   3.31MB / 737kB   11
af99b8cf271b   ca_orderer
```

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Step 11: Restart the Fabric Network

If network components are not responding correctly, restart the Fabric network.

```
./network.sh down

./network.sh up createChannel -ca
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.s
h down
./network.sh up createChannel -ca
Using docker and docker-compose
Stopping network
[+] down 10/10
 ✔Container ca_org1                    Removed                    3.6s
 ✔Container orderer.example.com        Removed                    0.5s
 ✔Container peer0.org2.example.com     Removed                    1.0s
 ✔Container ca_orderer                 Removed                    3.5s
 ✔Container peer0.org1.example.com     Removed                    1.0s
 ✔Container ca_org2                    Removed                    3.8s
 ✔Volume compose_peer0.org1.example.com Removed                   0.0s
 ✔Network fabric_test                  Removed                    0.2s
 ✔Volume compose_peer0.org2.example.com Removed                   0.0s
 ✔Volume compose_orderer.example.com   Removed                    0.0s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volum
e
Error response from daemon: get docker_peer0.org2.example.com: no such volum
e
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2
b5fbe664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd
3555f864cfa1398b52d341cb:latest
Deleted: sha256:35c754e2eb4e69f28dbdf3e440ad93aa7bc8e77e9660af6de414f1a32984
```

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Step 12: Verify the Network After Restart

Check the running containers and verify channel membership.

```
docker ps

peer channel list
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ docker ps
peer channel list
CONTAINER ID   IMAGE                               COMMAND
CREATED          STATUS           PORTS

                 NAMES
5d476e6f7f1e   hyperledger/fabric-peer:latest      "peer node start"
13 seconds ago   Up 13 seconds    0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp
, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
                 peer0.org2.example.com
327307710aa4   hyperledger/fabric-peer:latest      "peer node start"
13 seconds ago   Up 13 seconds    0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp
, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
                 peer0.org1.example.com
5844d84666d0   hyperledger/fabric-orderer:latest   "orderer"
13 seconds ago   Up 13 seconds    0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp
, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:
9443->9443/tcp   orderer.example.com
f656f6f97bbd   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"
20 seconds ago   Up 20 seconds    0.0.0.0:8054->8054/tcp, [::]:8054->8054/tcp
, 0.0.0.0:18054->18054/tcp, [::]:18054->18054/tcp
                 ca_org2
188c941aacae   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"
20 seconds ago   Up 20 seconds    0.0.0.0:7054->7054/tcp, [::]:7054->7054/tcp
, 0.0.0.0:17054->17054/tcp, [::]:17054->17054/tcp
                 ca_org1
```

Step 13: Clean Docker Resources

Optionally remove unused Docker resources.

```
docker system prune -f
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ docker syst
em prune -f
Deleted Containers:
0ebfc1ff5f56d347c5a930c7f22998d08108aed630de8035f49f4f9f9ac7b7b1
61277388e8e7fb30bc25d96fe24eaabda4e24095aaba74efa33a2969e2d8d932
0cbffbe4f6710afe5794470f98bb510025e183c83ffda9b31449c70e864a449e
ddb0e189193e62b60443cb8476a60124ac22963c5d9e781c03be87783f796afb
ac3fca05efed096cb053649f5bca11fd7bce98af420cc72c85ae037eed5499df
8f2567602d518eb1f1b66e01ff8ee5171e92a841167ad72a166051cb6cdbbe3d
6fc2e5c2458a860b0bda10b629154656b13d85d8d3b0712916423cbae9ebbcaa
da6b7b61bae16acba03280b416c156a49a2e271cc962433430e8263928d4acd9
```

COMMON ERRORS AND SOLUTIONS

Error	Possible Cause	Solution
Docker daemon not running	Docker service stopped	Start Docker using sudo systemctl start docker

Peer connection failed	Peer container stopped	Restart the Fabric network
Chaincode deployment failed	Incorrect chaincode path	Verify the -ccp directory
Channel not found	Channel was not created	Run ./network.sh createChannel
Access denied	Invalid MSP or certificates	Verify the MSP configuration and certificates
Chaincode not installed	Installation failed	Redeploy the chaincode
TLS handshake failed	Incorrect TLS certificate	Verify the TLS CA certificate paths
Port already in use	Another service is using the port	Stop the conflicting process

TROUBLESHOOTING WORKFLOW

Start Fabric Network ↓

Verify Docker Containers ↓

Check Peer Logs ↓

Check Orderer Logs ↓

Verify Channel ↓

Verify Chaincode ↓

Test Application ↓

Identify Error ↓

Apply Solution ↓

Restart Network if Required ↓

Verify Successful Execution

RESULT

The Hyperledger Fabric application was successfully analyzed, and common operational issues were identified and resolved using Docker commands, Fabric CLI tools, and log analysis. The experiment demonstrated effective troubleshooting techniques for network connectivity, chaincode deployment, channel configuration, certificate management, and resource monitoring.